

Software Engineering Coding Standards & Architecture Guidelines

1. Python Standards: Follow PEP 8 guidelines. All code must pass Black formatting, Flake8 linting, and MyPy type checking before PR submission.
2. Microservices Architecture: Services must communicate asynchronously via message queues or RESTful APIs with Pydantic contract validation.
3. Security Practices: Never commit hardcoded credentials or API keys. Use environment variables and secrets manager vaults.
4. Unit Testing: Maintain a minimum of 80% test coverage across core application logic. Integration tests must be automated in CI/CD.